Lecture 6.2: Structural Models

AREC 570 / ECON 530 · Unit 6

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Before class

Assigned

- Nevo and Whinston, “Taking the Dogma out of Econometrics: Structural Modeling and Credible Inference” (Nevo and Whinston 2010)
- Nevo, “A Practitioner’s Guide to Estimation of Random-Coefficients Logit Models of Demand” (optional) (Nevo 2000a)

Readings and course handouts are available on Canvas.

Also used in these slides

- Timmins and Schlenker, from Lecture 6.1 (Timmins and Schlenker 2009)
- Bellemare, on notation (Bellemare 2020)
- BLP demand: Berry (1994); Berry et al. (1995); Nevo (2001); Nevo (2000b)
- Conlon and Gortmaker, on best practices and PyBLP (Conlon and Gortmaker 2020)

Latin letters mark what you observe; Greek letters mark what you estimate

	Latin letters	Greek letters
Examples	y, x, W, P, V	α alpha, β beta, γ gamma, δ delta, ε epsilon
Used for	Variables you observe in the data	Parameters you estimate, and the error term you never observe

- Latin letters are the letters of the English alphabet
- Fixed effects are parameters, so they get Greek letters: γ_c, δ_t
- A Greek letter tells the reader the term is estimated

Source: Bellemare (2020), Section 5.1.

Subscripts tell the reader the level at which each term varies

$$y_{cst} = \beta P_{st} + \gamma_c + \delta_t + \varepsilon_{cst}$$

- One letter per dimension, used the same way throughout the paper: c county, s state, t year
- Order from the smallest unit to the largest, with time last
- A term carries only the subscripts it varies over: a state policy is P_{st}
- A fixed effect's subscript says what it holds constant: γ_c county, δ_t year
- The outcome's subscripts state the unit of observation: a county in a year

Reduced-form and structural approaches

A structural model recovers the parameters of behavior

- Definition: any empirical model that recovers the parameters of behavioral functions, such as demand and supply equations, utility functions, or cost structures
- It starts from a theoretical model and states what the researcher can and cannot observe
- What cannot be observed becomes a structural error, backed out of observed behavior
- The parameters are estimated with a procedure such as GMM or maximum likelihood

Source: Timmins and Schlenker (2009), Section 2.

The term “reduced form” has two meanings

- **Textbook meaning:** solve a system of equations so the outcome depends only on exogenous variables
- **Recent meaning:** program evaluation or treatment effects, with attention to how the exogenous variable was assigned
- Avoids as many structural assumptions as possible
- Reduced-form studies still require assumptions of their own

The designs from Lecture 6.1 are reduced form in the recent sense: fixed effects, quasi-experiments, regression discontinuity, and randomization.

Source: Timmins and Schlenker (2009), Section 3.

The two approaches differ in what they estimate and what they assume

	Reduced form	Structural
Estimates	High-level response parameters	Behavioral parameters, such as preferences and costs
Credibility comes from	The source of variation	The source of variation and the economic model
Best suited to	Establishing causality for a change in the data	Feedback loops, large policy changes, and welfare
Main assumptions	How treatment was assigned	Functional form, unobservables, and how agents behave (such as profit maximization)
Counterfactual	A comparison group, for a change in the data	A simulation of the model, including changes not in the data
Main limit	Harder to extend to changes not observed	Conclusions depend on the model being right

Source: Timmins and Schlenker (2009), Sections 1 and 4.

Structural models are used for feedback loops, new policies, and welfare

- **Feedback loops:** an outcome changes the conditions that produced it, so agents re-optimize and markets re-equilibrate
- **Large or new policies:** the change has not happened, or not at this scale
- **Welfare:** effects are converted to dollars through a utility function
- **Who bears the cost:** heterogeneity in preferences changes the distributional answer
- **Unintended consequences:** simulate the world with one piece of the policy turned off

Source: Timmins and Schlenker (2009), Section 2.1.

Structural analysis recovers parameters, then simulates counterfactuals

1. Match observed behavior to a theoretical model to recover parameters such as preferences and technology
2. Use the model to predict responses to changes, assuming those parameters are unchanged

Sources: Nevo and Whinston (2010), pages 70 to 71; Timmins and Schlenker (2009), Section 4.

Comparing reduced-form and structural approaches to merger analysis

	Reduced form: treatment effect of a past merger	Structural: merger simulation
Method	Compare prices near and far from the merging firms, before and after	Estimate demand and pricing, then simulate joint pricing
Strength	Clear identification for that merger	Predicts a proposed merger, with welfare
Limit	Other mergers may not be similar; no welfare measure	Holds products, firms, and pricing behavior fixed; instruments are often imperfect

Source: Nevo and Whinston (2010), pages 73 to 77.

Types of structural models you may see

Topic	Structural models	What they recover
Consumer demand	Demand systems (AIDS, QUAIDS); discrete choice (logit to BLP)	Elasticities, substitution, welfare
Nonmarket values	Hedonic; Ricardian; travel cost; choice experiments	Willingness to pay
Producers	Production, cost, and profit functions; stochastic frontier	Returns to scale, efficiency
Firms and markets	Conduct and markups; equilibrium displacement	Markups, pass-through, incidence
Dynamics and risk	Real options; dynamic discrete choice; risk preferences and insurance demand	Value of waiting, risk aversion
Trade and policy	Structural gravity; spatial equilibrium; CGE	Trade costs, economy-wide effects

Each method has an example specification on the methods map.

BLP: one structural model

BLP addresses the large number of cross-price effects in differentiated-product demand

A log-log demand system estimates one equation for each product j, with every price on the right side:

$$\ln q_{jt} = \gamma_j + \sum_{k=1}^{100} \eta_{jk} \ln p_{kt} + \varepsilon_{jt}$$

- η_{jk}: the percent change in the quantity of j when the price of k rises 1 percent
- With 100 products, a log-log demand system has at least 10,000 parameters
- BLP describes each product by its characteristics, so the number of parameters depends on the number of characteristics
- It uses market-level data on prices and quantities, and it allows for endogenous prices

Sources: Nevo (2000a), pages 514 to 515; Berry et al. (1995).

Nevo (2000): market concentration does not show how close brands are as substitutes

- **Empirical question:** how much would prices rise if two cereal firms merged?
- **Traditional merger analysis:** define the market, then compute a concentration measure, such as the HHI, from market shares
- **Limit:** with differentiated products the market is hard to define, and the HHI is a reasonable guide only if products are equally spaced in their characteristics
- **Portfolio effect:** a firm that owns two substitutes charges more than two separate firms would, because some sales lost by one brand go to the other
- **Nevo's approach:** estimate how much brands substitute, such as Post Grape Nuts and Nabisco Shredded Wheat, then simulate post-merger prices

Sources: Nevo (2000b); Nevo (2001), pages 307 to 308.

The theoretical model behind BLP: consumers, firms, and equilibrium

	Consumers	Firms
Choice	One product, or none	Prices of their own products
Objective	Maximize utility	Maximize profit: $\sum_{j \in F_f} (p_j - mc_j)\, Ms_j(p)$
Take as given	Income, products, and prices	Demand, marginal costs, and rivals' prices

- **Equilibrium:** Bertrand-Nash; each firm's prices are optimal given rivals' prices, and shares equal what consumers choose at those prices
- **Researcher observes:** market shares, prices, and product characteristics
- **Researcher does not observe:** unobserved quality, individual tastes and shocks, and marginal cost

Sources: Berry et al. (1995), Sections 2 and 3; Nevo (2001).

A structural section states the same four choices as any empirical section

Choice	In a structural paper
Model	The primitives: preferences, technology, and how agents interact
Estimation and inference	The estimator in one paragraph, with details in an appendix
Identification	Which variation identifies which parameter
Reported results	Fit or validation, and the counterfactuals the model is used for

Model: each consumer chooses the option that gives the highest utility

$$u_{ijt} = \alpha_i(y_i - p_{jt}) + x_{jt}\beta_i + \xi_{jt} + \varepsilon_{ijt}$$

- u_{ijt}: utility of consumer i from product j in market t (a city in a quarter)
- $y_i - p_{jt}$: income left after buying j; α_i: marginal utility of income
- x_{jt}: observed characteristics, such as sugar content
- ξ_{jt}: characteristics the researcher does not observe, such as brand quality or promotions
- β_i: tastes for those characteristics; α_i and β_i vary across consumers
- ε_{ijt}: a mean-zero shock
- Buying nothing is the outside good: $u_{i0t} = \alpha_i y_i + \varepsilon_{i0t}$

Summing individual choices gives each product's predicted market share. The data are the observed shares.

Source: Nevo (2000a), pages 517 to 520.

Logit and random-coefficients logit differ in whether tastes vary across consumers

	Logit	Random-coefficients logit (BLP)
Tastes	Same for everyone: α, β	Vary across consumers: α_i, β_i
How consumers differ	Only through the shock ε_{ijt}	Through tastes and the shock
Market shares	A closed-form formula	An average over consumers, computed numerically
Substitution	In proportion to market shares	Toward products with similar characteristics
Estimation	Linear instrumental-variables regression	GMM with a numerical search

Logit is the special case of the utility function above in which every consumer has the same tastes.

Source: Nevo (2000a), pages 522 to 526.

Estimation: back out the unobserved term from market shares, then use GMM

In logit, observed market shares convert into a linear regression:

$$\ln s_{jt} - \ln s_{0t} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

	ξ_{jt}: **unobserved product characteristics**	ε_{ijt}: **individual shock**
Varies by	Product and market	Consumer, product, and market
Known to	Consumers and firms, so it affects price	The consumer only
In estimation	The error term in the regression	Averaged over consumers into the share formula

- **Random coefficients:** for each guess of the taste parameters, find the ξ_{jt} that reproduce observed shares, and choose the guess that makes ξ_{jt} uncorrelated with the instruments (GMM)
- In a paper, this gets one paragraph; the steps go in an appendix (derivation notes)

Sources: Berry (1994); Nevo (2000a), pages 522 and 530 to 533.

BLP identification has two parts: the price coefficient and substitution patterns

	Price coefficient	Substitution patterns
Parameters	Mean price sensitivity, α	How tastes vary across consumers
Problem	Endogeneity: price is correlated with ξ_{jt}	Market shares are aggregates, so they do not show who switched to what
What identifies it	Instruments that shift price	How shares respond across products and markets
Threats	Instruments correlated with demand shocks	Too little variation in prices, characteristics, choice sets, or demographics

Only the first part is an endogeneity problem. The second concerns which variation pins down the spread of tastes.

Source: Nevo (2000a), pages 528 to 530.

Identification of the price coefficient: price is endogenous, so it needs instruments

- Price is an equilibrium outcome: firms set markups knowing demand, so p_{jt} is correlated with ξ_{jt}
- OLS biases the price coefficient toward zero, because high-ξ_{jt} products have both higher prices and higher shares
- Identification requires instruments Z_{jt} that shift price and satisfy $E[\xi_{jt} \mid Z_{jt}] = 0$

Instrument	Why it moves price	Exclusion restriction
Cost shifters	Shift marginal cost	Uncorrelated with demand shocks
Characteristics of rival products (BLP)	Close rivals lower markups	Product characteristics are set before demand shocks are realized
Prices of the same brand in other markets (Hausman)	Same brand shares marginal cost across markets	Demand shocks are independent across markets

Source: Nevo (2000a), pages 528 and 534 to 536.

Random coefficients: Π and Σ measure how tastes vary across consumers

Each consumer's taste for a characteristic, here sugar, has three parts:

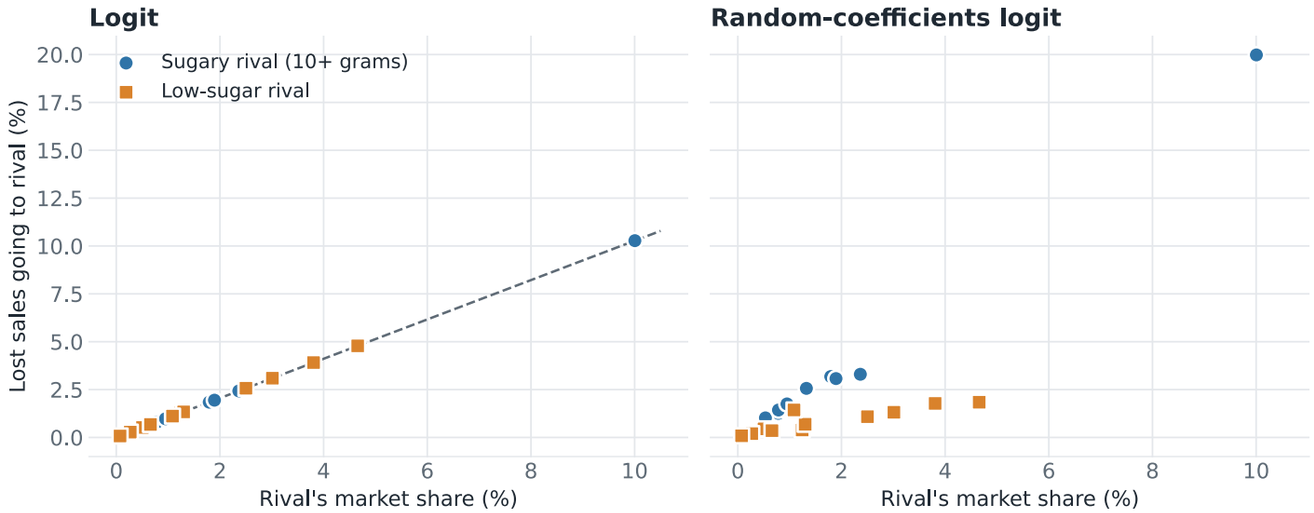
$$\beta_i^{\text{sugar}} = \beta^{\text{sugar}} + \pi_{\text{inc}}\,\text{income}_i + \pi_{\text{age}}\,\text{age}_i + \sigma\,\nu_i$$

Parameter	What it measures	Demo estimate (SE)
β^{sugar}	Taste of the average consumer	In product fixed effects
π_{inc}, π_{age} (in Π)	How the taste changes with income and age	−0.385 (0.122), 0.052 (0.026)
σ (in Σ)	Spread of the taste among consumers with the same income and age	0.006 (0.014)

- Stacking these rows for price and every characteristic gives $\Pi D_i + \Sigma \nu_i$; logit sets $\Pi = 0$ and $\Sigma = 0$
- In the demo, lower-income consumers like sugar more, so sugary cereals share the same buyers

Source: Nevo (2000a), pages 519 and 526.

Demo: with random coefficients, more of a sugary cereal's lost sales go to sugary rivals



- Each point is a rival cereal. Further right is a larger market share; higher is a larger part of the lost sales
- **Logit:** the points form a line, so market share alone sets a rival's gain
- **Random coefficients:** sugary rivals, which share buyers, gain more than low-sugar rivals of the same size

A cereal with 14 grams of sugar, one market: sugary rivals get 22.7% in logit, 40.5% with random coefficients.

Identification of substitution patterns: compare how shares respond across products and markets

Nevo's example: products A and B are similar, and B and C have the same market share. The price of A rises.

- **Logit predicts** equal changes in the shares of B and C
- **If B gains more than C**, the data favor random coefficients, and the size of the gap identifies the parameters in Σ
- **Demographics:** the parameters in Π are identified by comparing markets with different demographics

Having many markets helps when they add variation in prices, characteristics, choice sets, and demographics. Nevo's cereal data have no variation in the choice set across markets, which he offers as a reason his estimates of Σ are imprecise.

Source: Nevo (2000a), pages 529 to 530 and 541.

Results: a structural paper reports demand estimates, then simulation results

	Demand estimates	Simulation results
Comes from	Estimating the model on observed data	Solving the estimated model under a new scenario
Reports	Parameters and standard errors; elasticities; substitution; implied markups	Prices, quantities, profits, and consumer welfare under the scenario
Check	Are elasticities and markups sensible?	What is held fixed? Does the model predict well out of sample?
In BLP (1995)	Tables IV to VIII	Not simulated; Section 8 lists uses such as merger and trade policy

The demand estimates are also a result in their own right. A paper can stop there.

Sources: Berry et al. (1995), Sections 7 and 8; Nevo and Whinston (2010), pages 75 to 77.

Simulation results: a merger simulation has four steps

1. Start from the demand estimates
2. Recover marginal costs from firms' pricing conditions: $p - mc = \Omega^{-1}s(p)$
3. Change ownership: the merged firm sets the prices of both firms' products
4. Solve for new equilibrium prices, then compute shares and welfare

- State what is held fixed: current merger simulations hold products, firms, and pricing behavior fixed
- Report results under a range of assumptions
- Validate the model: check that it predicts outcomes not used to estimate it

Sources: Nevo (2001); Nevo and Whinston (2010), pages 75 to 77; Timmins and Schlenker (2009), Section 2.

Demo: cereal results

Demo: estimating cereal demand in steps, then simulating a merger

- **Data:** Nevo's simulated cereal data, 24 products in 94 markets (47 cities, 2 quarters)
- Nevo: the data are "not real and should not be used for any analysis," so the numbers here illustrate the method
- **Demand estimates:** logit by OLS, add product fixed effects, add instruments, add random coefficients
- **Simulation:** a merger of firms 1 and 2
- **Software:** PyBLP in Python; the script runs in under a minute

Open the cereal demand demo

Sources: Nevo (2000a), page 538; Conlon and Gortmaker (2020).

Demand estimates: product fixed effects triple the price coefficient, because quality was omitted

Specification (all three are logit)	Price coefficient
OLS, no product fixed effects	−10.1
OLS, product fixed effects	−29.0
Instrumental variables, product fixed effects	−30.0

- Better cereals cost more, so without fixed effects a high price partly signals high quality
- Product fixed effects absorb a product's fixed quality, which is part of ξ_{jt}
- **The instruments are for price.** In Nevo (2001), they are prices of the same brand in other cities and cost proxies
- Random coefficients come next; these rows address the price coefficient only

Demand estimates: random coefficients change substitution far more than own-price elasticities

	Logit	Random-coefficients logit
Mean price coefficient	−30.0	−62.7
Median own-price elasticity	−3.65	−3.61
Largest rival response ÷ smallest	1.0	6.2
Median margin	31.6%	33.7%

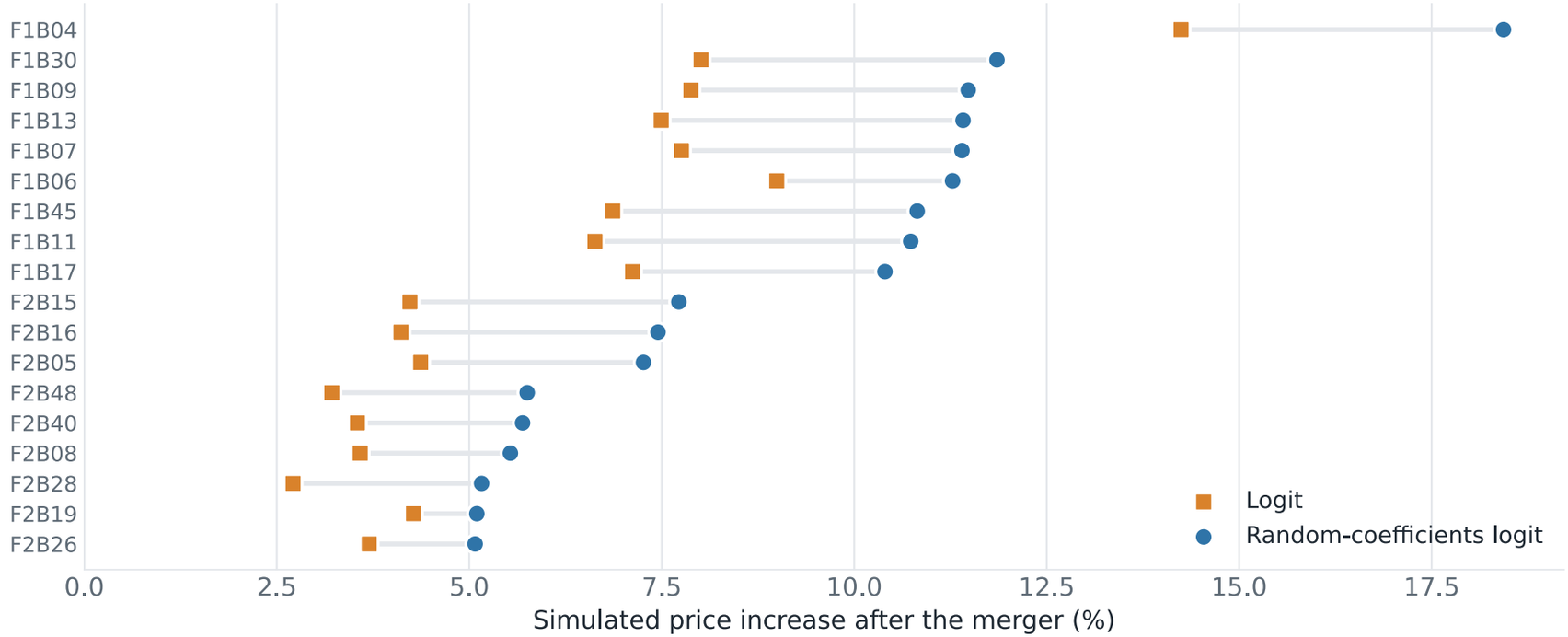
- Elasticities and margins are what the estimates imply; the demo page shows each calculation
- The second row barely moves, and the third row moves a lot

Simulation results: two models with the same own-price elasticity give different merger predictions

Merger of firms 1 and 2	Logit	Random-coefficients logit
Median price change, merging firms	5.8%	11.6%
Median price change, other firms	0.1%	0.4%
Change in consumer surplus	−11.6%	−13.6%

The counterfactual depends on how the model lets consumers substitute, which is an assumption the section has to state and defend.

Simulation results: the predicted increase is higher for every merging product



One market of the demo. Median predicted increase: 5.5% in logit and 9.1% with random coefficients.

Using a structural model

A structural section reports five items

1. The model's primitives
2. Which variation identifies which parameter
3. The estimator in one paragraph, with details in an appendix
4. How well the model fits or validates
5. The counterfactuals it is used for

This is row 10 of the empirical strategy checklist.

Numerical estimation can stop early, so report how the model was solved

Estimation searches for the parameters that minimize the GMM objective. The optimizer stops when it is close enough, and that stopping rule is a choice.

Same model, same data	Mean price coefficient	Mean own-price elasticity
Loose stopping rule (1E-4), as published	−32.4	−3.70
Tighter stopping rule (1E-5), as in our demo	−62.7	−3.62
Tighter stopping rule and better instruments	−27.5	−3.69

- **How to read this:** the coefficient moves a lot, and the elasticity barely moves
- Report the optimizer, tolerances, starting values, and integration method
- Try several optimizers and starting values, and report the quantities that matter

Source: Conlon and Gortmaker (2020), Table 7.

Pairing reduced-form evidence with a structural model is good practice

- A credible design shows that the response exists in the data
- Reduced-form estimates can serve as inputs to the model or as targets it must reproduce
- Retrospective studies check the accuracy of model predictions
- Reporting why the approaches agree or disagree gives the reader more confidence than either one alone

Sources: Timmins and Schlenker (2009), Sections 1 and 4; Nevo and Whinston (2010), page 77.

What to think about when using a structural model

- **The question:** does it require a model? Name the reason: feedback, a new policy, welfare, or distribution
- **The parameters:** which ones does the model recover, and what variation identifies each one?
- **The assumptions:** which ones drive the counterfactual, and how do results change under alternatives?
- **The reduced-form evidence:** what credible estimate supports the model?
- **The validation:** does the model predict outcomes not used to estimate it?

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Course navigation

- All lectures
- Schedule and preparation
- Course reading list
- Empirical strategy checklist
- Methods map
- BLP derivation notes
- Cereal demand demo
- Download this lecture as a PDF